

The Use of R Packages for Big Data Analytics

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R is a powerful statistical programming language for data science. It consists of powerful packages to deal with Big Data analytics, which is a complex process of examining large and varied data sets in order to reveal hidden, useful, novel and interesting trends and patterns. The R programming language comprises packages and environments for statistical computing with Big Data, making analytics easier.



The R programming language is the right choice for Big Data analytics as it provides capabilities such as access to parallel and distributed machine learning algorithms as well as parallel data and task execution. R programming supports both explicit and implicit parallelism to handle Big Data effortlessly. This language has a rich set of packages for Big Data analytics.

Hadoop is a well known framework for the distributed processing of large data sets. R provides the capabilities of Hadoop with its RHadoop based packages. RHadoop is a collection of five main packages related to distributed processing in R. These sets of packages handle basic connectivity to the Hadoop distributed file system (HDFS) and to the HBASE distributed database. They

perform common data manipulation operations, statistical analysis using Hadoop MapReduce functionality and read-write operations on *arvo* files from the local and HDFS file systems.

ORCH is another vital package of R that deals with Big Data. ORCH is a collection of several packages that consist of interfaces to deal with Hive tables, the local R environment and Oracle database tables. These sets of packages are very efficient for predictive Big Data analytics.

RHIPE is another vital package of R for Big Data analytics. This package provides an integrated environment of R and Hadoop for Big Data processing.

RHadoop package

RHadoop is a collection of the five packages listed in Table 1.

To install RHadoop, follow the steps given below:

- i. Download Hortonworks Sandbox for VMware.
- ii. Install VMware Player.
- iii. Open VMware, navigate to Hortonworks Sandbox and import.
- iv. Install RStudio in the sandbox.
- v. To log into the virtual machine, click into your virtual machine and type `<Alt + F5>`. Use the default user name: *root* and password: *Hadoop*
- vi. Run the following commands:

```
sudo rpm -Uvh
sudo yum -y install git wget R
ls/etc/default
sudo ln -s /etc/default/hadoop /etc/
profile.d/hadoop.sh
```